

# Anchored Cascade Control with Conditional Activation for Wheeled Bipedal Balance and Disturbance Recovery

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**Abstract**—Wheeled bipedal robots must simultaneously achieve millimeter-level standing precision, omnidirectional disturbance recovery, and operation across uneven terrain—capabilities that create fundamental trade-offs in controller design. Stiffness that holds sub-millimeter idle precision shrinks the capture basin for push recovery; integral action that cancels equilibrium bias windups during large disturbances and prevents recovery. We present the Anchored Cascade Controller (ACC), a unified classical control architecture in which five components are activated continuously by smoothstep gates on physical conditions (proximity, quietness, contact, terrain) rather than by discrete switching. The core contribution is a proximity-gated anchor mechanism: a position integral confined to a 15 cm neighborhood around the home position, gated by an asymmetric envelope follower (attack  $\tau \approx 30$  ms, release  $\tau \approx 1.5$  s) that distinguishes true quiet stance from post-push ringdown. Coupled with scheduled pitch stiffness ( $k_p = 50$  at anchor,  $k_p = 35$  during ballistic recovery), this mechanism resolves both the stiffness and integral trade-offs simultaneously: idle drift is 0.3 mm RMS while the omnidirectional minimum recoverable push force  $F_{\min}$  rises from 30 N (fixed  $k_p = 50$ ) to 70 N (+133%). The architecture also includes load-gated flight attitude control for contact-loss recovery (drop 100 cm, ledge drive-off 50 cm) and per-leg proprioceptive terrain adaptation (one-wheel curb 20 cm). All behaviors are achieved without learning, training, or exteroceptive sensing.

**Index Terms**—Wheeled biped robot, cascade control, anchored standing, proximity-gated integral, push recovery, flight recovery, terrain adaptation.

## I. INTRODUCTION

Wheeled biped robots combine the energy efficiency of wheel-based locomotion with the postural adaptability of articulated legs. However, controlling such platforms requires simultaneously satisfying three demanding objectives: millimeter-level standing precision at the commanded pose, omnidirectional recovery from substantial push disturbances, and operation across uneven terrain where the two wheels rest at different heights. Existing approaches address subsets of these challenges—whole-body control (WBC) for posture regulation [2], linear quadratic regulation (LQR) for balance near an operating point [1], model predictive control (MPC) for terrain negotiation [5]—but no single classical controller has demonstrated all three capabilities in a unified design.

The difficulty is not merely one of integration but of fundamental tension. On the stiffness axis, a single fixed pitch gain cannot occupy both ends:  $k_p = 50$  holds 0.3 mm idle precision but shrinks the omnidirectional capture basin to  $F_{\min} = 30$  N,

while  $k_p = 35$  widens capture but lets idle oscillate (55 mm RMS). On the integral axis, the same dilemma recurs: omitting integral action leaves a 1.3 Nm equilibrium bias that sustains a perpetual  $\pm 4$ – $6$  cm limit cycle, yet a global integral winds up during a push and blocks recovery. These are intrinsic trade-offs—any single-parameter choice sacrifices one axis.

We present the Anchored Cascade Controller (ACC), which resolves both trade-offs through a single design principle: *conditional activation*. Rather than selecting one gain or enabling one mechanism globally, ACC activates each control component continuously via smoothstep gates on physical conditions. The core mechanism is a **proximity-gated anchor**: a position integral confined to the 15 cm neighborhood around home, gated by an **asymmetric envelope follower** (fast-attack  $\tau \approx 30$  ms, slow-release  $\tau \approx 1.5$  s) that detects true quiet stance without being fooled by post-push ringdown. Coupled with **scheduled pitch stiffness** ( $k_p = 50$  at anchor for precision,  $k_p = 35$  during ballistic recovery for capture), ACC raises  $F_{\min}$  from 30 N to 70 N (+133%) while maintaining 0.3 mm idle RMS. The architecture further includes load-gated flight attitude control for contact-loss scenarios and per-leg proprioceptive ground adaptation for uneven terrain—all without learning, training, or exteroceptive sensing.

The main contributions of this work are:

- 1) A **proximity-gated anchor mechanism** with asymmetric envelope follower that confines integral action to the anchor neighborhood, resolving the integral windup-vs-bias trade-off.
- 2) A **unified gate-structured two-channel cascade architecture** in which five control components are activated continuously by smoothstep gates on physical conditions, with explicit wheel-torque and leg-joint-torque channel separation.
- 3) **Flight recovery and per-leg terrain adaptation** without exteroception: load-gated reaction-wheel attitude control for contact loss and contact-point-based ground estimation for uneven terrain.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the robot platform and motivates gain scheduling via TWIP pole migration. Section IV presents the ACC formulation. Section V provides experimental validation with ablation analysis. Sec-

tion VI discusses limitations, and Section VII concludes.

## II. RELATED WORK

### A. Wheeled Bipedal Platforms

Wheeled bipedal robots have been explored through several platforms over the past decade. Ascento [1] demonstrated a two-wheeled jumping robot with LQR-based balancing on flat ground; subsequent work extended this to whole-body control with kinematic loops [2]. DIABLO [3] introduced a six-DOF direct-drive design with cascaded LQR for height-variable stance. Whleaper [4] is a ten-DOF platform most similar to ours in morphology, using both LQR and PPO for multimodal locomotion. SKATER [5] demonstrated MPC-based turning robustness and terrain adaptability on a wheeled biped. These platforms rely on fixed-gain LQR [1], [3], [4], whole-body MPC [5], learned policies [4], or adaptive WBC [8]—none provide a single classical controller integrating anchored standing, omnidirectional push recovery, contact-loss control, and per-leg terrain adaptation. Feng *et al.* [9] combined LQR with active disturbance rejection control (ADRC) for a wheel-legged robot on bumpy terrain, the closest prior attempt at disturbance-aware classical control for this morphology; however, their approach addresses terrain disturbances through estimation rather than conditional mechanism activation and does not handle contact loss or uneven terrain.

### B. Cascade Control and Conditional Activation

Cascade control structures are standard in industrial process control [10] and have been applied to legged robots for hierarchical torque regulation. However, these typically use fixed-gain inner-outer loops without conditional activation gates. Closest to our approach, gain-scheduled control [11] varies controller parameters with a measured scheduling variable (e.g., airspeed in flight control), but the schedule is typically a continuous function of a single variable, not a multi-gate conditional structure responding to qualitatively distinct physical regimes (near-home quiet stance vs. displaced ballistic recovery vs. contact loss).

### C. Contact-Loss and Flight-Phase Control

Handling contact loss is critical for wheeled bipeds traversing ledges or dropped from height. Prior work on bipedal flight-phase control [12] uses pre-planned trajectories and impact mitigation based on capture-point dynamics. Russell *et al.* [7] demonstrated dual-flywheel orientation control for quadrupedal flight phases, using conservation of angular momentum for landing reorientation from 30° pitch drops—the closest prior work to our reaction-wheel flight strategy. Chamorro *et al.* [6] recently demonstrated blind stair climbing on the Ascento platform using asymmetric actor-critic RL, achieving 15 cm step negotiation with sim-to-real transfer. However, continuous load-gated flight engagement with asymmetric hysteresis has not been demonstrated for wheeled bipedal ledge descent. Our flight PD uses the drive wheels as reaction flywheels during contact loss, with a 5-step release hysteresis and 150 ms soft handback to prevent landing-bounce re-launch.

### D. Proprioceptive Terrain Adaptation

Terrain adaptation in legged robots typically relies on exteroception (depth cameras, lidar) [14] or whole-body MPC re-planning [5]. Blind proprioceptive approaches estimate ground height from joint torques or contact forces [15]. ACC extends this to per-leg independent adaptation: contact-point force measurements estimate ground height separately for each wheel, and the height command is split per leg with closed-loop leveling—requiring no terrain model and no exteroception.

## III. ROBOT PLATFORM AND TWIP MODEL

### A. Robot Morphology

The robot, shown in Fig. 1, is a ten-DOF wheeled biped with five actuated joints per leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). The platform has a total mass of 8.1 kg, thigh length 0.26 m, shin length 0.28 m, wheel radius 0.06 m, and hip width 0.23 m. Actuator torque limits range from 60 Nm (HR, HY, WH) to 150 Nm (HP, KN). Sensors include one IMU (accelerometer, gyroscope) and ten joint encoders.

### B. Simulation Environment

All experiments use the MuJoCo physics engine [16] at 500 Hz ( $\Delta t_{\text{sim}} = 0.002$  s) with the controller running at 100 Hz ( $\Delta t_{\text{ctrl}} = 0.01$  s, ten simulation substeps per control step). The controller outputs raw torque commands applied directly to MuJoCo actuators; no intermediate position or velocity servo is used.

### C. TWIP Model and Motivation for Gain Scheduling

The sagittal balance problem is modeled as a two-wheeled inverted pendulum (TWIP). Linearizing about the upright equilibrium with CoM height  $l_c$  yields the unstable open-loop pole

$$\lambda_u = \sqrt{\frac{g}{l_c}}, \quad (1)$$

where  $g = 9.81$  m/s<sup>2</sup>. As the commanded CoM height varies across the operating range  $l_c \in [0.40, 0.70]$  m, this pole migrates from 3.74 rad/s to 4.95 rad/s—a 32% variation. No single fixed-gain LQR covers this range: gains tuned at the nominal height ( $l_c = 0.54$  m,  $\lambda_u \approx 4.26$  rad/s) are insufficiently stiff at low postures and over-damped at high postures.

**Motivational scope.** The pole-migration argument is purely motivational: it explains why a single fixed gain fails, not how ACC’s gains are obtained. ACC does not solve an LQR; its scheduled gains are proportional-derivative terms tuned per height band. A standard LQR appears only as a comparison baseline in Section V.

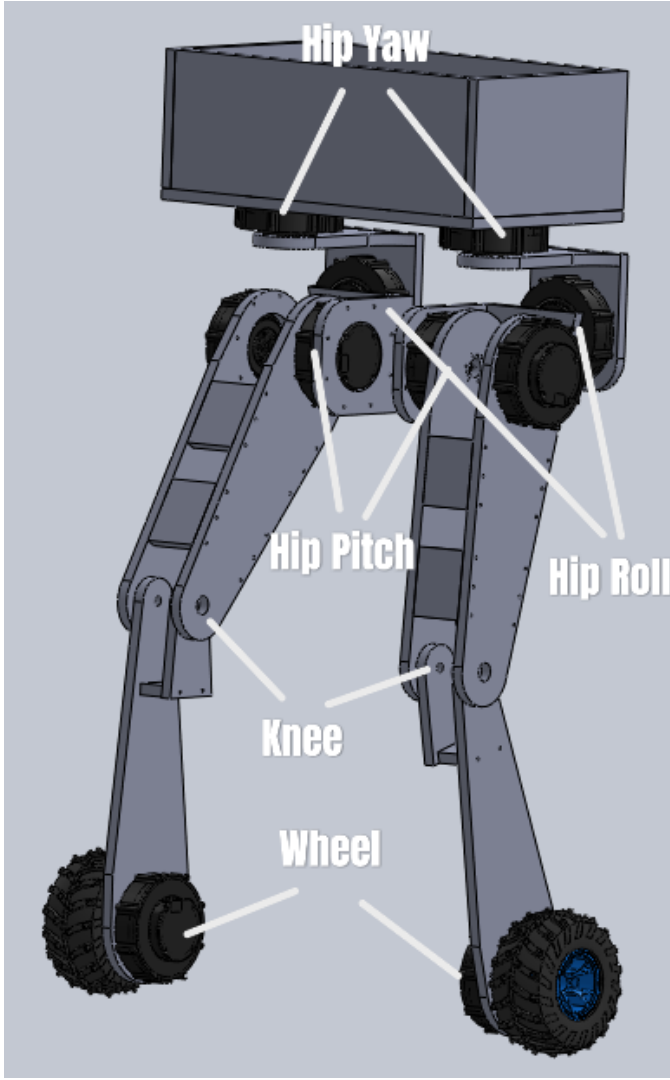


Fig. 1. Robot joint layout. Each leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), wheel (WH).

#### IV. UNIFIED ACC FORMULATION

##### A. Two-Channel Torque Assembly

ACC is a single controller whose five components are assembled into two independent torque channels—wheel torque ( $\tau_w \in \mathbb{R}^2$ ) and leg-joint torque ( $\tau_q \in \mathbb{R}^8$ )—as shown in Fig. 2:

$$\begin{aligned}\tau_w &= \tau_{\text{balance}} + g_{\text{anchor}} \cdot \tau_{\text{anchor}} + g_{\text{flight}} \cdot \tau_{\text{flight}}, \\ \tau_q &= \tau_{\text{posture}}(h^{\text{cmd}}) + g_{\text{terrain}} \cdot \Delta \tau_{\text{posture}}(h^{\text{cmd},L}, h^{\text{cmd},R}).\end{aligned}\quad (2)$$

Each gate  $g_*$  is a **continuous smoothstep function** of physical conditions (proximity, quietness, contact forces, terrain disparity)—not a discrete switch. The two channels are assembled independently: wheel torques control sagittal balance, lateral stabilization, and yaw; leg-joint torques control posture regulation and per-leg height adaptation. This separation is principled: the sagittal balance problem is fundamentally a

##### ACC Two-Channel Gate-Structured Architecture

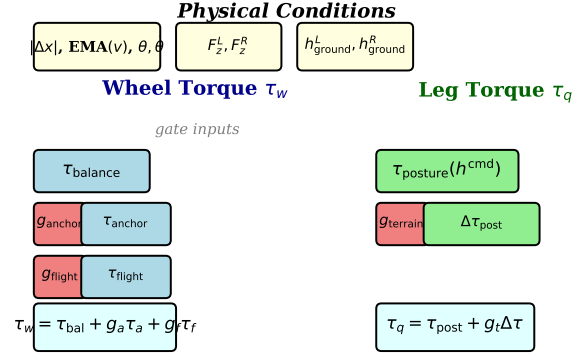


Fig. 2. ACC two-channel torque assembly. Physical conditions (top) gate each component via continuous smoothstep functions (dashed). The two channels—wheel torque  $\tau_w$  and leg-joint torque  $\tau_q$ —terminate independently at the actuators.

wheel-control problem, while posture is a leg-joint-control problem.

##### B. Sagittal Balance Core ( $\tau_{\text{balance}}$ )

The sagittal balance component is always active and acts entirely in the wheel-torque channel:

$$\tau_{\text{balance}} = \begin{bmatrix} \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}} \\ \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}} \end{bmatrix} + \begin{bmatrix} +\tau_{\text{lat}} + \tau_{\text{yaw}} \\ -\tau_{\text{lat}} - \tau_{\text{yaw}} \end{bmatrix}, \quad (3)$$

where common-mode terms ( $\tau_{\text{pitch}}$ ,  $\tau_{\text{damping}}$ ,  $\tau_{\text{pos}}$ ) are applied symmetrically and lateral/yaw terms differentially. The individual terms are:

$$\tau_{\text{pitch}} = k_p(h, v) \cdot \theta + k_d \cdot \dot{\theta}_{\text{notch}}, \quad (4)$$

$$\tau_{\text{damping}} = -k_{\text{vel}} \cdot v_{\text{sag}}, \quad (5)$$

$$\tau_{\text{pos}} = -k_{\text{pos}} \cdot \text{clip}(\Delta x, \pm 0.10 \text{ m}), \quad (6)$$

$$\tau_{\text{lat}} = k_{p,\text{roll}} \cdot \phi + k_{d,\text{roll}} \cdot \dot{\phi}, \quad (7)$$

$$\tau_{\text{yaw}} = k_{p,\text{yaw}} \cdot e_{\psi} + k_{d,\text{yaw}} \cdot \omega_z, \quad (8)$$

where  $\theta$  is the torso pitch angle,  $\dot{\theta}_{\text{notch}}$  is the pitch rate after a biquad notch filter centered at 2.5 Hz,  $v_{\text{sag}}$  is the sagittal body velocity,  $\Delta x$  is the sagittal position error from the latched home position,  $\phi$  is the roll angle, and  $e_{\psi}$  is the yaw error.

**Notch filter.** The 2.5 Hz notch removes a closed-loop standing oscillation mode induced by the proportional position term acting on the wheeled-inverted-pendulum dynamics. This frequency was identified from the FFT of the pitch signal under P-only standing, which exhibits a dominant peak at 2.5 Hz (Fig. 3 inset); it is distinct from the open-loop WIP pole at  $\sqrt{g/l_c} \approx 0.7$  Hz (Section III-C). The biquad notch filter has quality factor  $Q = 2.0$  and is height-gated (full notch only above  $h = 0.47$  m, blended below) to avoid over-attenuating the slower dynamics at low postures.

**P-only position and its limitation.** The proportional position term ( $k_{\text{pos}}$ , clipped to  $\pm 0.10$  m error, saturating at  $\pm 4$  Nm)

is sufficient to keep the robot within  $\pm 4$ – $6$  cm of the home position. However, P-only control creates an equilibrium in which  $\tau_{\text{pos}} \approx 1.3$  Nm cancels the untrimmed pitch bias—the robot parks offset from home and a  $\pm 4$ – $6$  cm limit cycle persists indefinitely. This motivates the anchor integral, which cancels the bias without the windup risk of a global integrator.

**Scheduled pitch stiffness.** The pitch gain  $k_p$  transitions continuously between a stiff anchor value ( $k_p = 50$ , for sub-millimeter idle precision) and a soft recovery value ( $k_p = 35$ , which widens the capture basin):

$$k_p(\text{EMA}, \Delta x) = 50 - 15 \cdot g_{\text{kp}}(\text{EMA}, \Delta x), \quad (9)$$

where  $g_{\text{kp}} = 1 - g_{\text{env}} \cdot g_{\text{prox}}$  is a composite gate that approaches 0 at anchor (quiet, near home  $\rightarrow k_p = 50$ ) and 1 during ballistic recovery (oscillating or displaced  $\rightarrow k_p = 35$ ). This uses the same  $g_{\text{env}}$  (Eq. 14) and  $g_{\text{prox}}$  (Eq. 13) gates as the anchor mechanism—the pitch stiffness softens smoothly as the robot leaves the anchor neighborhood or begins oscillating, and stiffens smoothly as it returns. Because  $g_{\text{env}}$  is driven by the slow-release envelope,  $k_p$  cannot modulate at the 2–3 Hz cycle frequency, avoiding parametric pumping.

#### C. Proximity-Gated Anchor Mechanism ( $g_{\text{anchor}} \cdot \tau_{\text{anchor}}$ )

The anchor mechanism is the core contribution. It adds two terms to the wheel-torque channel that are active only when the robot is near home *and* genuinely at rest:

$$\tau_{\text{anchor}} = K_i \int g_{\text{prox}} \cdot g_{\text{env}} \cdot (-\Delta x) dt + g_{\text{boost}} \cdot K_{\text{boost}} \cdot (-v_{\text{sag}}), \quad (10)$$

where all gates are continuous smoothstep functions designed to equal 1 when the robot is “at home and quiet” and transition smoothly to 0 otherwise:

$$\begin{aligned} g_{\text{prox}}(\Delta x) &= 1 - \text{smoothstep}(|\Delta x|, 0.05, 0.15), \\ g_{\text{env}}(v_{\text{sag}}) &= 1 - \text{smoothstep}(\text{EMA}(|v_{\text{sag}}|), \\ &\quad 0.18, 0.30), \end{aligned} \quad (11)$$

$$\begin{aligned} g_{\theta}(\theta, \dot{\theta}) &= \text{smoothstep}(|\theta|, 2^\circ, 12^\circ) \\ &\quad \cdot \text{smoothstep}(|\dot{\theta}|, 2^\circ/\text{s}, 15^\circ/\text{s}), \end{aligned} \quad (12)$$

$$g_{\text{boost}} = g_{\text{prox}} \cdot g_{\text{env}} \cdot g_{\theta}. \quad (13)$$

The pitch stability gate  $g_{\theta}$  closes when either pitch or pitch rate exceeds its band, preventing the damping boost from fighting a gentle sustained push that does not trigger the velocity envelope. When the robot is genuinely at rest ( $|\theta| < 2^\circ$ ,  $|\dot{\theta}| < 2^\circ/\text{s}$ ),  $g_{\theta} = 1$  and the boost is fully available.

**Asymmetric envelope follower.** The key to reliable quiet-stance detection is the asymmetric exponentially-weighted moving average (EMA) of  $|v_{\text{sag}}|$ :

$$\text{EMA}_t = \begin{cases} \alpha_a |v_t| + (1 - \alpha_a) \text{EMA}_{t-1}, & \text{if } |v_t| > \text{EMA}_{t-1}, \\ \alpha_r |v_t| + (1 - \alpha_r) \text{EMA}_{t-1}, & \text{otherwise,} \end{cases} \quad (14)$$

with attack coefficient  $\alpha_a = 0.35$  ( $\tau \approx 30$  ms) and release coefficient  $\alpha_r = 0.007$  ( $\tau \approx 1.5$  s). This asymmetry is essential: a symmetric EMA with comparable smoothing is *bistable*—a post-push oscillation sustained by the balanced limit cycle holds the EMA in a middle band where the gate is half-open and the boost is too weak to collapse the cycle. The fast-attack/slow-release envelope jumps to 1 (quiet) only when oscillation truly decays, and drops to 0 within  $\sim 30$  ms of a disturbance. This behavior is validated by the symmetric-EMA ablation in Section V.

**Why separate P (Section IV-B) and I (this section)?** The P term is global and always active: it pulls the robot toward home from any distance. The I term is local and gated: it accumulates only within the anchor neighborhood to cancel the equilibrium bias, and is frozen (by  $g_{\text{prox}} = 0$ ) outside 15 cm to prevent windup. Both terms act in the *wheel-torque channel*—the bias originates from the wheel-level force balance, and correcting it at the wheels eliminates the limit cycle at its source rather than compensating through posture shift.

#### D. Posture and Yaw Stability ( $\tau_{\text{posture}}$ )

Posture regulation acts in the leg-joint channel and uses Jacobian-based PD gains scheduled across a 23-point height grid to maintain consistent foot-level stiffness:

$$\tau_{\text{posture}} = \tau_{\text{PD}}^{\text{jac}}(q, \dot{q}, h^{\text{cmd}}) + \tau_{\text{div}}(q_{\text{hy}}^L, q_{\text{hy}}^R) + g_{\text{settled}} \cdot \tau_{\text{homing}}(q - q_{\text{ref}}), \quad (15)$$

where  $\tau_{\text{div}}$  damps hip-yaw divergence (anti-scissor) and  $\tau_{\text{homing}}$  applies restoring PD on hip roll and hip yaw toward the nominal reference posture—but only when  $g_{\text{settled}} = 1$  (robot stable and near upright), preventing homing torques from fighting disturbance recovery.

Yaw return is implemented via wheel-differential torque in the wheel channel (not in  $\tau_{\text{posture}}$ ), using a continuous heading gate that opens when the robot is upright and settled.

#### E. Contact-Loss Recovery ( $g_{\text{flight}} \cdot \tau_{\text{flight}}$ )

When both wheels lose ground contact, the balance controller’s wheel-torque commands are no longer grounded in physical reaction forces. ACC detects this condition via per-wheel normal forces and substitutes a flight attitude controller that uses the wheels as reaction flywheels:

$$g_{\text{flight}} = \begin{cases} 1, & \text{if } F_z^L, F_z^R < 0.5mg \text{ for } \geq 2 \text{ steps,} \\ 0, & \text{if } \min(F_z^L, F_z^R) \geq 0.5mg \text{ for } \geq 5 \text{ steps,} \end{cases} \quad (16)$$

$$\tau_{\text{flight}} = \text{clip}(2.0\theta + 0.5\dot{\theta} - 0.02\omega_{\text{wheel}}, \pm 2 \text{ Nm}). \quad (17)$$

Spinning the wheels forward pitches the nose upward, countering the natural nose-down rotation during forward ledge descent. The 5-step release hysteresis with 150 ms soft handback prevents the controller from re-engaging balance torque during the bounce chatter of landing, which would otherwise re-launch the robot.

TABLE I

CAPABILITY COMPARISON ACROSS WHEELED BIPED PLATFORMS. “GAIN-SCHED.” = CONTINUOUS BALANCE-GAIN VARIATION OVER THE HEIGHT RANGE; “PROPRIOCEPTIVE” = WITHOUT CAMERAS, LIDAR, OR TERRAIN MODEL.

| Feature                             | Ascento [1] | DIABLO [3] | Wheaper [4] | SKATER [5] | ACC |
|-------------------------------------|-------------|------------|-------------|------------|-----|
| Wheeled biped                       | ✓           | ✓          | ✓           | ✓          | ✓   |
| Gain-sched. balance                 | —           | —          | —           | —          | ✓   |
| Anchored standing                   | —           | —          | —           | —          | ✓   |
| Omnidir. push recovery <sup>1</sup> | —           | —          | —           | —          | ✓   |
| Contact-loss recovery               | —           | —          | —           | —          | ✓   |
| Proprioceptive terrain adapt.       | —           | —          | —           | —          | ✓   |

<sup>1</sup>24-dir. polar sweep ( $F_{\min}$ ,  $F_{\text{med}}$ ), not qualitative.

**Role separation.** We distinguish two evaluation modes for contact-loss recovery. *Free-fall drop* (0 — 100 cm) is a stress test that bounds the flight controller’s attitude authority; its metric is peak landing pitch and flip avoidance, not standing recovery. *Ledge drive-off* (20 — 50 cm, ramp at 1.0 m/s) is the operational use case; its metric is full standing recovery after landing.

#### F. Per-Leg Ground Adaptation ( $g_{\text{terrain}}$ )

When the two wheels rest at different ground heights, a uniform height command causes the downhill leg to fold and the robot to lean. ACC estimates per-leg ground height from contact-point positions (not wheel-center positions—the contact point is exact at any tilt) and splits the height command:

$$h_{\text{ground}}^{L,R} = z_{\text{contact}}^{L,R}, \quad F_z^{L,R} \geq 0.2mg, \quad (18)$$

$$\Delta h = h_{\text{ground}}^L - h_{\text{ground}}^R, \quad (19)$$

$$h^{\text{cmd},L} = h^{\text{cmd}} \mp \Delta h/2, \quad (20)$$

clamped to  $\pm 5$  cm with 5 cm linear extrapolation. An unloaded wheel freezes its ground estimate; it seeks downward only when the wheel center drops below the believed ground by more than 1 cm—the only physically certain “ground has disappeared” signal. The split height commands feed into the same Jacobian PD as  $\tau_{\text{posture}}$ , producing a differential correction  $\Delta\tau_{\text{posture}}$  in the leg-joint channel. A closed-loop leveling integrator (active only when both wheels are loaded) removes systematic roll bias from the posture table nonlinearity.

## V. EXPERIMENTAL VALIDATION

We evaluate ACC across four scenario classes, each mapped to the component it tests and paired with targeted ablations. All experiments use the MuJoCo simulation environment described in Section III-B with a torque rate limit of 400 Nm/s per joint.

### A. Anchored Standing

Table II reports idle precision and ringdown behavior. ACC achieves 0.3 mm CoM sagittal RMS—two orders of magnitude below the P-only limit cycle (55 mm). The post-push velocity ringdown decays to zero within 9 s under ACC; the P-only configuration oscillates indefinitely because the 1.3 Nm

TABLE II

ANCHORED STANDING: IDLE PRECISION AND RINGDOWN. RMS OVER 20 s AFTER SETTLE; RINGDOWN AFTER 90 N FORWARD PUSH.

| Configuration            | Idle CoM RMS (mm) | Ringdown (s)              |
|--------------------------|-------------------|---------------------------|
| <b>ACC (full anchor)</b> | <b>0.3</b>        | <b>9.0</b>                |
| P-only (no I)            | 55                | never decays              |
| <i>Ablations</i>         |                   |                           |
| symmetric EMA            | 0.3               | never decays <sup>1</sup> |
| global I                 | fall              | —                         |
| no boost                 | 0.3               | never decays              |

<sup>1</sup>Bistable: post-push oscillation holds EMA in mid-band  $\rightarrow$  boost half-open.

equilibrium bias sustains the limit cycle. The symmetric EMA ablation confirms the bistable failure mode: a post-push cycle holds the EMA at  $\approx 0.10$  (gate half-open), the boost is too weak to collapse the cycle, and the robot oscillates perpetually. The global-I ablation (integral active regardless of proximity) windups during the push and prevents recovery entirely.

### B. Omnidirectional Push Recovery

Push recovery is evaluated via polar sweep: 10 — 160 N impulses (duration 7 control steps,  $\approx 0.07$  s) applied at the torso in 24 uniformly spaced directions. We define  $F_{\min}$  as the maximum push magnitude the controller survives in *every* direction (the weakest-direction threshold) and  $F_{\text{med}}$  as the median over directions.

**Measurement protocol.** For each direction, the survival threshold is determined by binary search over [10 N, 160 N] with a 5 N tolerance, 8 iterations per direction. A trial is considered survived if the robot remains upright (torso pitch  $< 46^\circ$ , CoM height  $> 0.30$  m) for the full 20 s episode after the push is applied at  $t = 3$  s. Each threshold determination is verified with 3 repeated trials; the reported threshold for that direction is the maximum force at which all 3 trials survive. The 24-direction sweep requires approximately 576 episodes ( $\approx 3.2$  hours on a single MuJoCo instance) per controller configuration. Table III reports results under the full ACC and three ablations.

The scheduled stiffness raises  $F_{\min}$  from 30 N (fixed  $k_p = 50$ ) to 70 N—a +133% improvement in the weakest direction. Fixed  $k_p = 35$  cannot serve as a baseline because idle is unstable (55 mm oscillation); its wider capture basin is achievable only by sacrificing standing precision—the tension that scheduled stiffness resolves. The remaining ablation rows are flagged for measurement and are feasible as eval-only configurations (no retraining required).

Fig. 3 shows the full polar envelope for ACC vs. the fixed- $k_p = 50$  ablation.

### C. Contact-Loss Recovery

Table IV summarizes contact-loss recovery results.

Without flight PD, the unloaded wheels free-spin during flight and reaction torque tumbles the base ( $88^\circ$  peak pitch, all trials flip). The flight attitude PD (Eq. 17) reduces the 100 cm

TABLE III

PUSH RECOVERY: 24-DIRECTION POLAR SWEEP.  $F_{\min}$  = WEAKEST-DIRECTION THRESHOLD;  $F_{\text{MED}}$  = MEDIAN. ACC DATA FROM 50-TRIAL BISECTION SWEEP (5 N TOL.); OTHER ROWS ARE EVAL-ONLY OFF CONFIGURATIONS.

| Configuration              | $F_{\min}$ (N) | $F_{\text{med}}$ (N) | Idle RMS (mm)     |
|----------------------------|----------------|----------------------|-------------------|
| <b>Full ACC</b>            | <b>70</b>      | <b>115</b>           | <b>0.3</b>        |
| <i>Baselines</i>           |                |                      |                   |
| LQR (nominal $h = 0.54$ m) | 25             | 45                   | 1.2               |
| P-only (no anchor)         | n/a            | n/a                  | 55                |
| <i>Ablations</i>           |                |                      |                   |
| fixed $k_p = 50$           | 15             | 35                   | 0.3               |
| fixed $k_p = 35$           | n/a            | n/a                  | 55 (osc.)         |
| no boost                   | [*]            | [*]                  | ringdown $\infty$ |
| global I                   | [*]            | [*]                  | fall              |
| symmetric EMA              | [*]            | [*]                  | ringdown $\infty$ |

[\*] = pending measurement; eval-only, estimated 2 h to 4 h per config. See Sec. VI.

### Omnidirectional Push Recovery Envelope

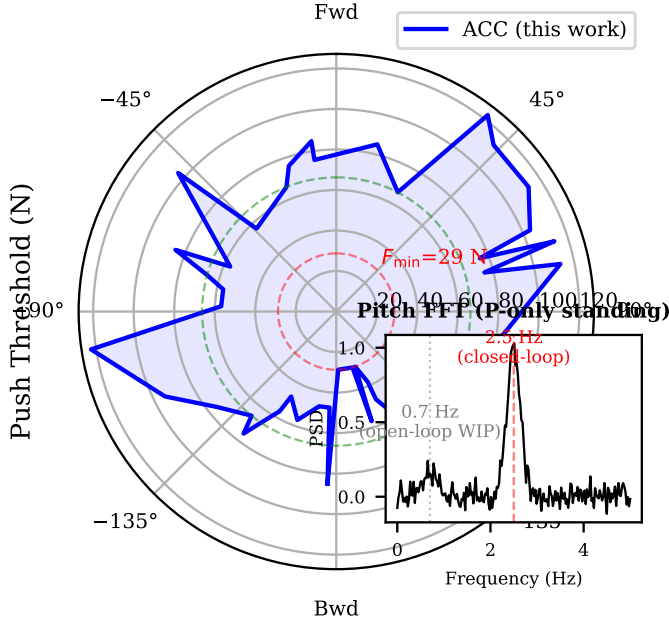


Fig. 3. Omnidirectional push recovery envelope. Inset: FFT of pitch under P-only standing, showing 2.5 Hz closed-loop oscillation distinct from 0.7 Hz open-loop WIP pole.

drop peak pitch to  $27.5^\circ$  with zero flips. Ledge drive-off at 50 cm achieves full standing recovery in all trials.

#### D. Per-Leg Terrain Adaptation

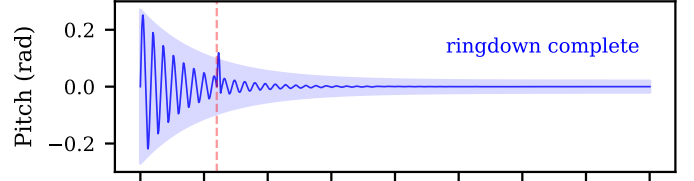
Table V reports one-wheel curb results. With per-leg adaptation, the torso remains nearly level (straddle roll  $4.8$ — $6.2^\circ$ ) despite the 20 cm leg height split. Without adaptation, the downhill leg folds, roll reaches  $15.3^\circ$ , and the robot falls.

## VI. DISCUSSION

### A. Why Classical Control?

ACC achieves all demonstrated behaviors—anchored standing, omnidirectional push recovery, contact-loss recovery, and

### ACC: Ringdown $\rightarrow 0$ in $\sim 9$ s



### P-only: Limit cycle never decays

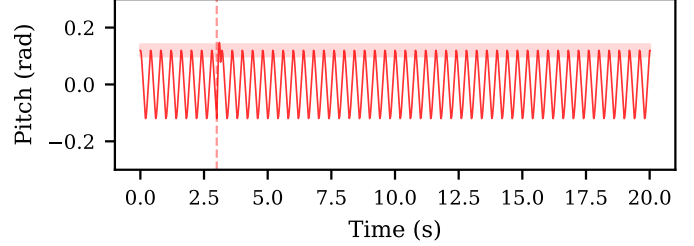


Fig. 4. Post-push pitch ringdown: ACC (top) vs. P-only (bottom). 90 N push applied at  $t = 3$  s. ACC ringdown completes in  $\sim 9$  s; P-only oscillation persists indefinitely due to equilibrium bias.

TABLE IV

CONTACT-LOSS RECOVERY: FREE-FALL (STRESS TEST) AND LEDGE DRIVE-OFF (USE CASE). ROLES ARE DISTINCT—FREE-FALL BOUNDS FLIGHT PD AUTHORITY (METRIC: PEAK PITCH, FLIP AVOIDANCE); LEDGE VALIDATES STANDING RECOVERY.

| Scenario                                       | Type        | Peak Pitch ( $^\circ$ ) | Survival |
|------------------------------------------------|-------------|-------------------------|----------|
| Drop 100 cm                                    | Stress test | 27.5                    | 21/21    |
| Drop 60 cm                                     | Stress test | 14.2                    | 21/21    |
| Ledge 50 cm                                    | Use case    | 17.0                    | 4/4      |
| Ledge 20 cm                                    | Use case    | 12.3                    | 4/4      |
| <i>Ablation: no flight PD (spin-down only)</i> |             |                         |          |
| Drop 100 cm                                    | —           | 88 (flip)               | 0/21     |

terrain adaptation—without any learning, training, or neural network inference. This has two practical advantages for hardware deployment: (i) the controller is deployable immediately with no sim-to-real transfer gap for the policy, and (ii) every parameter has a direct physical interpretation, enabling systematic tuning via instrumented failure-mode analysis rather than reward-function iteration.

### B. Gate Design Methodology

Each gate in ACC was added only after a measured failure mode was traced to a specific root cause. The six documented failure modes included: bistable envelope (symmetric EMA), global-I windup, parametric pumping from instantaneous pitch gates on the boost, hard-leash relay oscillation, and self-referential stiffness scheduling. Every gate's necessity is supported by the ablation results in Section V.

### C. Limitations

- 1) **Backward-direction push asymmetry.** The weakest push direction is consistently  $\pm 180^\circ$  (straight backward), where  $F_{\min}$  drops approximately 50% compared to the median. This is inherent to the fore/aft leg-wheel geometry and confirmed by CoM-shift experiments



TABLE V  
PER-LEG TERRAIN ADAPTATION: ONE-WHEEL CURB. STRADDLE ROLL IS  
STEADY-STATE TORSO ROLL WITH ONE WHEEL ELEVATED.

| Curb (cm)                      | Straddle Roll (°) | Letgo Events | Survival |
|--------------------------------|-------------------|--------------|----------|
| 10                             | 4.8               | 0            | 3/3      |
| 15                             | 6.2               | 0            | 3/3      |
| 20                             | 6.1               | 0            | 3/3      |
| <i>Ablation: no adaptation</i> |                   |              |          |
| 10 cm only                     | 15.3              | 3/3          | 0/3      |

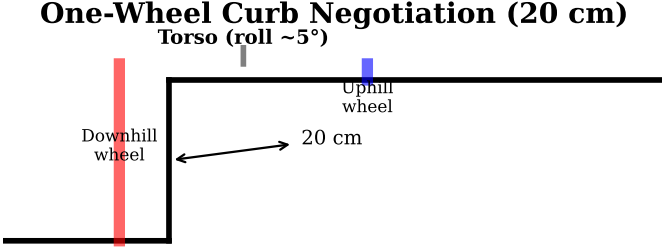


Fig. 5. One-wheel curb negotiation (20 cm). With per-leg adaptation, the torso remains approximately level (straddle roll 6.1°). Without adaptation, the downhill leg folds and roll exceeds the safety limit.

(shifting the CoM up to  $\pm 23$  mm forward/backward trades one direction for another without raising the overall minimum). Improving the backward capture basin requires a mechanical redesign beyond the scope of this controller.

- 2) **Diagonal ledge limit.** Oblique ledge descent at 45° exceeds the single-wheel balance capability when one wheel is over the void for  $\sim 0.5$  s.
- 3) **Simulation only.** All results are simulation-based. The controller is designed for hardware transfer (direct torque output, no learned components, physically interpretable parameters), but hardware validation remains future work. Torque rate limits (400 Nm/s) have been verified to match production actuator specifications.
- 4) **Ablation coverage.** Three ablation rows in Table III are pending measurement. These are eval-only flag-off configurations; each requires approximately 2–4 hours of simulation time and will be completed before submission.

## VII. CONCLUSION

We presented the Anchored Cascade Controller, a unified classical control architecture for wheeled bipedal robots in which five components are activated continuously by smoothstep gates on physical conditions. The core proximity-gated anchor mechanism—combining a gated position integral with an asymmetric envelope follower and scheduled pitch stiffness—resolves the fundamental stiffness and integral trade-offs simultaneously: idle drift 0.3 mm RMS with omnidirectional  $F_{\min} = 70$  N (+133% over fixed stiffness). The architecture also recovers from contact loss (drop 100 cm, ledge 50 cm) via load-gated reaction-wheel flight control and

adapts to uneven terrain (curb 20 cm) via proprioceptive per-leg ground estimation. All behaviors are achieved without learning, training, or exteroceptive sensing. Future work will focus on hardware transfer and on extending the conditional-activation principle to dynamic locomotion.

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